

Set Number: Set-1

Question COS (a)

Question: Define with example of replenishment cost, procurement cost, and stock-out-cost related to inventory control.

Marks: 3

Solution:

- **Replenishment Cost (Ordering Cost):** This is the cost incurred every time an order is placed to replenish inventory. It includes fixed costs associated with the order process, such as administrative expenses, paperwork, transportation, and receiving/inspection costs, regardless of the order size.
 - *Example:* A company pays a flat shipping and handling fee of \$150 every time they order a new batch of raw materials, whether they order 10 units or 10,000 units.
- **Procurement Cost (Purchase Cost):** This is the direct, actual purchasing price of the items being bought to add to the inventory.
 - *Example:* If a hardware store buys 50 hammers from a supplier at a wholesale price of \$12 each, the procurement cost is \$600.
- **Stock-out Cost (Shortage Cost):** This is the economic penalty incurred when an item is demanded but not available in inventory. It can include the cost of lost sales, emergency expedited shipping, backorder administration, and the intangible loss of customer goodwill.
 - *Example:* A car dealership doesn't have a specific brake pad in stock. They lose a \$100 service job to a competitor because the customer refuses to wait for the part.

Question COS (b)

Question: A shopkeeper has a uniform demand of an item at the rate of 50 units per month. He buys from supplier at the cost of Rs. 6/- per item and the cost of ordering is Rs. 10/- each time. If the stock holding costs are 20% per year of stock value, how frequently should he replenish his stocks? What is the optimal cost of inventory and what is the total cost?

Marks: 3

Solution:

First, we must standardize all units to a yearly basis.

- **Annual Demand (D):**

$$50 \text{ units/month} \times 12 \text{ months/year} = 600 \text{ units/year}$$

- **Unit Cost (C):** Rs. 6/item

- **Ordering Cost (S):** Rs. 10/order

- **Holding Cost (H):**

$$20\% \text{ of unit cost per year} = 0.20 \times 6 = \text{Rs. } 1.2 \text{ /unit/year}$$

1. Optimal Order Quantity (Economic Order Quantity - EOQ):

We use the EOQ formula to find the optimal lot size (Q^*):

$$Q^* = \sqrt{\frac{2DS}{H}}$$

$$Q^* = \sqrt{\frac{2 \times 600 \times 10}{1.2}} = \sqrt{\frac{12000}{1.2}} = \sqrt{10000} = 100 \text{ units}$$

2. Frequency of Replenishment (t^*):

The time between orders is the order quantity divided by the annual demand.

$$t^* = \frac{Q^*}{D} = \frac{100}{600} = \frac{1}{6} \text{ year}$$

Since $\frac{1}{6}$ of a year is 2 months ($12 \div 6 = 2$), the shopkeeper should replenish his stocks every 2 months (or 6 times a year).

3. Optimal Cost of Inventory (Total Variable Cost - TVC):

This consists of the annual ordering cost plus the annual holding cost.

$$TVC = \text{Ordering Cost} + \text{Holding Cost} = \left(\frac{D}{Q^*}\right)S + \left(\frac{Q^*}{2}\right)H$$

$$TVC = \left(\frac{600}{100}\right)10 + \left(\frac{100}{2}\right)1.2 = (6 \times 10) + (50 \times 1.2) = 60 + 60 = \text{Rs. 120 per year}$$

4. Total Cost (TC):

Total cost is the sum of the optimal inventory cost and the actual purchasing cost of the materials.

$$TC = TVC + (D \times C)$$

$$TC = 120 + (600 \times 6) = 120 + 3600 = \text{Rs. 3720 per year}$$

Question COS (c)

Question: Define payoff matrix and saddle point. Apply max-min and min-max principle to solve the game (pay-off matrix $J_i \rightarrow M_j$):

Marks: 2+5 = 7

Solution:

Definitions:

- **Payoff Matrix:** A mathematical table that displays the possible outcomes (payoffs) of a game for each player based on the combination of strategies they choose. For a zero-sum game, the matrix typically shows the gains of the row player (and corresponding losses for the column player).
- **Saddle Point:** In game theory, a saddle point occurs when the maximum of the row minimums (Max-min) is exactly equal to the minimum of the column maximums (Min-max). It represents the optimal pure strategy for both players, and the payoff at this point is the "value of the game."

Applying the Max-Min and Min-Max Principle:

We find the minimum value in each row (Row Min) and the maximum value in each column (Col Max).

	M1	M2	M3	M4	M5	Row Min
J1	4	2	1	7	-1	-1
J2	1	-4	-6	-7	6	-7
J3	3	2	3	4	2	2 (Max-Min)
J4	-6	1	-1	0	4	-6
J5	0	0	6	0	0	0
Col Max	4	2 (Min-Max)	6	7	6	

- **Max-min Value:** The maximum of the row minimums is $\max(-1, -7, 2, -6, 0) = 2$.
- **Min-max Value:** The minimum of the column maximums is $\min(4, 2, 6, 7, 6) = 2$.

Since the **Max-min value (2) = Min-max value (2)**, a saddle point exists.

- **Optimal Strategy for Player J:** Strategy J_3
- **Optimal Strategy for Player M:** Strategy M_2
- **Value of the Game:** 2

Question C06 (a)

Question: At a public telephone booth in a post office arrivals are considered to be Poisson with an average inter-arrival time of 12 minutes. The length of phone call may be assumed to be distributed exponentially with an average of 4 minutes.

- What is the probability that a fresh arrival will not have to wait for phone?
- What is the average length of queues that form time to time?
- What is the fraction of time that the phone is busy?

Marks: 2+2+2 = 6

Solution:

This is an M/M/1 queuing model problem.

- **Arrival rate (λ):** $\frac{1}{12}$ arrivals per minute
- **Service rate (μ):** $\frac{1}{4}$ services per minute
- **Utilization factor / Traffic intensity (ρ):** $\frac{\lambda}{\mu} = \frac{1/12}{1/4} = \frac{4}{12} = \frac{1}{3}$

i) Probability that a fresh arrival will not have to wait:

This is the probability that the system is completely empty (idle).

$$P_0 = 1 - \rho = 1 - \frac{1}{3} = \frac{2}{3} \approx 66.67\%$$

ii) Average length of queues that form time to time:

The average length of the queue (L_q) across all time periods is:

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{(1/3)^2}{1 - (1/3)} = \frac{1/9}{2/3} = \frac{1}{6} \approx 0.167 \text{ customers}$$

(Note: If the question specifically implies the expected length of the queue strictly when a queue is actually present/non-empty, the formula is

$$\frac{\mu}{\mu - \lambda} = \frac{1/4}{1/4 - 1/12} = 1.5 \text{ customers}).$$

iii) Fraction of time that the phone is busy:

This is equal to the utilization factor (ρ).

$$\text{Busy Fraction} = \rho = \frac{1}{3} \approx 33.33\%$$

Question C06 (b)

Question: Define with example of optimistic time, pessimistic time, and slack time.

Marks: 3

Solution:

- **Optimistic Time (t_o):** The minimum possible time required to complete a project activity, assuming everything goes perfectly without any delays or setbacks.
 - *Example:* Pouring a concrete foundation might take 3 days under perfect weather conditions and with no equipment failures.
- **Pessimistic Time (t_p):** The maximum possible time required to complete a project activity, assuming the worst-case scenario where everything that can reasonably go wrong does go wrong (excluding catastrophic acts of God).
 - *Example:* The same concrete foundation might take 9 days if there are severe rainstorms, material delivery delays, and unexpected labor shortages.
- **Slack Time (or Float):** The amount of time that a task in a project network can be delayed without causing a delay to the subsequent tasks or to the overall project completion time. It is calculated as Late Start - Early Start ($LS - ES$) or Late Finish - Early Finish ($LF - EF$).
 - *Example:* If a project must be finished in 30 days, and a specific painting task can be completed anywhere between day 10 and day 15 without pushing the project past day 30, the slack time is 5 days.

Question C06 (c)

Question: A project consists of the following activities is given below.

- Find the earliest, latest and slack time of each activity.
- Hence find the critical path and project duration.

Marks: 8

Solution:

To find these times, we perform a Forward Pass to find Earliest Start (ES) and Earliest Finish (EF), followed by a Backward Pass to find Latest Start (

LS) and Latest Finish (LF).

- **Forward Pass:** $EF = ES + \text{Duration}$
- **Backward Pass:** $LS = LF - \text{Duration}$
- **Slack Time:** $LS - ES$ (or $LF - EF$)

Step i) Calculations for each activity:

Activity	Duration	Predecessor	ES	EF	LS	LF	Slack (LS-ES)
A	3	--	0	3	0	3	0
B	5	A	3	8	3	8	0
C	4	A	3	7	7	11	4
D	6	B	8	14	8	14	0
E	3	B	8	11	11	14	3
F	5	C	7	12	11	16	4
G	4	D, E	14	18	14	18	0
H	2	F	12	14	16	18	4
I	3	G, H	18	21	18	21	0

Step ii) Critical Path and Project Duration:

- The **Critical Path** consists of the sequence of activities that have a Slack Time of 0. Tracing the zero-slack activities from the table gives us: **A → B → D → G → I**
- The **Project Duration** is the Earliest Finish (EF) time of the final activity on the critical path, which is **21 days**.

Set Number: Set-2

Here are the detailed solutions for the examination paper.

Question COS (a)

Question: Give a graphical interpretation of EOQ. Measure the EOQ without shortage and deterministic uniform demand.

Marks: 3

Solution:

Graphical Interpretation of EOQ:

The Economic Order Quantity (EOQ) model can be graphically represented by plotting cost against order quantity.

1. **Holding Cost Curve:** This is a straight line starting from the origin and sloping upwards, representing that holding costs increase linearly as the order quantity increases.
2. **Ordering Cost Curve:** This is a hyperbola sloping downwards, showing that as the order quantity increases, the number of orders per year decreases, thereby reducing the total annual ordering cost.
3. **Total Cost Curve:** This is a U-shaped curve representing the sum of the holding and ordering costs.

The EOQ occurs at the lowest point (the trough) of the Total Cost curve. Mathematically and graphically, this minimum point is exactly where the Holding Cost curve intersects the Ordering Cost curve.

Measuring EOQ (Derivation):

Let:

- D = Annual demand rate (deterministic and uniform)
- S = Ordering cost per order
- H = Holding cost per unit per year
- Q = Order quantity

The total annual variable cost (TVC) is the sum of annual ordering cost and annual holding cost:

$$TVC = \left(\frac{D}{Q}\right)S + \left(\frac{Q}{2}\right)H$$

To find the minimum cost (EOQ), we differentiate TVC with respect to Q and set it to zero:

$$\frac{d(TVC)}{dQ} = -\frac{DS}{Q^2} + \frac{H}{2} = 0$$

$$\frac{DS}{Q^2} = \frac{H}{2}$$

$$Q^2 = \frac{2DS}{H}$$

$$EOQ (Q^*) = \sqrt{\frac{2DS}{H}}$$

Question COS (b)

Question: A stockiest has to supply 400 units of a product every Monday to his customers. He gets the product at Rs. 50/- per unit from the manufacturer. The cost of ordering and transportation from the manufacturer is Rs. 75 per order. The cost of carrying inventory is 7.5% per year of the cost of the product. Determine the economic lot size and the total optimal cost (including the capital cost).

Marks: 3

Solution:

First, standardise all variables to an annual basis (assuming 52 weeks in a year).

- **Annual Demand (D):** 400 units/week $\times$ 52 weeks/year = 20,800 units/year
- **Unit Cost (C):** Rs. 50/unit
- **Ordering Cost (S):** Rs. 75/order
- **Holding Cost (H):** 7.5% of Rs. 50 = $0.075 \times 50 =$ Rs. 3.75/unit/year

1. **Economic Lot Size (EOQ):**

$$EOQ = \sqrt{\frac{2DS}{H}}$$

$$EOQ = \sqrt{\frac{2 \times 20800 \times 75}{3.75}} = \sqrt{\frac{3120000}{3.75}} = \sqrt{832000} \approx \mathbf{912.14 \text{ units}}$$

2. Total Optimal Cost:

Total cost includes the variable inventory costs (holding + ordering) plus the capital cost of purchasing the units ($D \times C$).

At EOQ, Total Variable Cost (TVC) is calculated as:

$$TVC = \sqrt{2DSH} = \sqrt{2 \times 20800 \times 75 \times 3.75} = \sqrt{11700000} \approx \text{Rs. } 3420.53$$

$$\text{Total Cost (TC)} = TVC + (D \times C)$$

$$TC = 3420.53 + (20800 \times 50)$$

$$TC = 3420.53 + 1040000 = \mathbf{\text{Rs. } 1,043,420.53 \text{ per year}}$$

Question COS (c)

Question: How will you solve a game without any saddle point? For what value of λ , the game is strictly determinable.

Pay-off matrix:

	B1	B2	B3
A1	λ	7	3
A2	-2	λ	-8
A3	-3	4	λ

Marks: 2+5 = 7

Solution:

1. Solving a game without a saddle point:

Games without a saddle point are solved using **Mixed Strategies**. Instead of choosing a single optimal strategy, players assign probabilities to their available

choices. Methods to solve this include:

- **Algebraic / Probability Method:** Equating the expected payoffs for each strategy to find the probability distribution.
- **Graphical Method:** Applicable for $2 \times n$ or $m \times 2$ matrices to reduce them to 2×2 matrices.
- **Linear Programming:** For larger $m \times n$ matrices, the game is converted into primal and dual linear programming problems to find optimal expected values.

2. Value of λ for strict determinability:

A game is strictly determinable if a saddle point exists, meaning **Max-min = Min-max**.

Let's evaluate the Row Minimums:

- Row A_1 : $\min(\lambda, 7, 3) = \min(\lambda, 3)$
- Row A_2 : $\min(-2, \lambda, -8) \leq -2$ (Specifically, it will be -8 if $\lambda \geq -8$, or λ if $\lambda < -8$)
- Row A_3 : $\min(-3, 4, \lambda) \leq -3$

The Max-min value is the maximum of these row minimums. Since rows A_2 and A_3 yield negative values, the Max-min is dictated by Row 1:

$$\text{Max-min} = \min(\lambda, 3)$$

Let's evaluate the Column Maximums:

- Col B_1 : $\max(\lambda, -2, -3) = \max(\lambda, -2)$
- Col B_2 : $\max(7, \lambda, 4) = \max(\lambda, 7)$
- Col B_3 : $\max(3, -8, \lambda) = \max(\lambda, 3)$

The Min-max value is the minimum of these column maximums. Because Col 2 is at least 7, the Min-max must be the lesser of Col 1 or Col 3:

$$\text{Min-max} = \min(\max(\lambda, -2), \max(\lambda, 3))$$

For a saddle point to exist:

$$\text{Max-min} = \text{Min-max}$$

$$\min(\lambda, 3) = \min(\max(\lambda, -2), \max(\lambda, 3))$$

If we test the interval $-2 \leq \lambda \leq 3$:

- Row 1 minimum is λ .
- Column 1 maximum is λ , Column 3 maximum is 3. The minimum of the column maximums is λ .
- Therefore, $\text{Max-min} = \text{Min-max} = \lambda$. The game is strictly determinable for $-2 \leq \lambda \leq 3$.

Question C06 (a)

Question: Customers arrive at one window drive in a bank according to a Poisson distribution with a mean of 10 per hour. Service time per customer is exponential with a mean of 5 minutes. The space in front of the window, including that for the serviced car can accommodate a maximum of three cars. Other cars can wait outside the space.

- i) How long an arriving customer is expected to wait before starting service?
- ii) What is the probability that an arriving customer can drive directly to the space in front of the window and will have to wait outside the indicated space?

Marks: $2+2+2 = 6$

Solution:

This is an M/M/1 queue problem.

- Arrival rate (λ): 10 customers/hour
- Service time: 5 minutes = $5/60$ hour = $1/12$ hour.
- Service rate (μ): 12 customers/hour
- Traffic Intensity (ρ): $\frac{\lambda}{\mu} = \frac{10}{12} = \frac{5}{6}$

i) Expected waiting time before service (W_q):

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{5/6}{12 - 10} = \frac{5/6}{2} = \frac{5}{12} \text{ hours}$$

To convert to minutes: $\frac{5}{12} \times 60 = \mathbf{25 \text{ minutes}}$.

ii) Probabilities for space allocation:

The prompt asks for two distinct probabilities combined in one sentence:

- **Probability a customer can drive directly to the space:** This means there are fewer than 3 cars already in the system ($N < 3$).

$$P(N < 3) = P_0 + P_1 + P_2$$

Using the formula $P_n = (1 - \rho)\rho^n$, the sum is $1 - \rho^3$.

$$P(N < 3) = 1 - \left(\frac{5}{6}\right)^3 = 1 - \frac{125}{216} = \frac{91}{216} \approx \mathbf{0.4213 \text{ (or 42.13\%)}}$$

- **Probability a customer will have to wait outside the space:** This means the space is full, so there are 3 or more cars already in the system ($N \geq 3$).

$$P(N \geq 3) = 1 - P(N < 3) = \left(\frac{5}{6}\right)^3 = \frac{125}{216} \approx \mathbf{0.5787 \text{ (or 57.87\%)}}$$

Question C06 (b)

Question: How the CPM type of networks differ from the PERT networks?

What is crash time?

Marks: 3

Solution:

Differences between CPM and PERT:

1. **Time Estimates:** CPM uses a deterministic approach with a single, highly accurate time estimate for each activity. PERT uses a probabilistic approach

with three time estimates (Optimistic, Pessimistic, and Most Likely) to calculate an expected time.

2. **Orientation:** CPM is activity-oriented (focusing on the tasks themselves), whereas PERT is event-oriented (focusing on the start and completion milestones).
3. **Application:** CPM is generally used for repetitive, predictable projects (like construction), while PERT is used for unpredictable, non-repetitive projects (like R&D).
4. **Cost vs. Time:** CPM incorporates cost-time trade-offs (crashing), while PERT is strictly a time-centric model.

Crash Time:

Crash time is the absolute minimum possible duration in which an activity can be completed. Achieving crash time typically requires allocating additional resources (extra labor, overtime, specialized equipment), which invariably results in a higher cost (known as crash cost).

Question C06 (c)

Question: A small project is composed of 7 activities whose time estimates are listed below.

- i) Find the expected project completed time.
- ii) Calculate the probability that the project will be completed at least 3 weeks than expected.
- iii) If the project due date is 18 weeks, what is the probability of not meeting the due date?

Marks: 8

Solution:

First, we calculate the expected time (t_e) and variance (σ^2) for each activity.

Formulas: $t_e = \frac{t_o + 4t_l + t_p}{6}$ and $\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$

Activity (i-j)	t_o	t_l	t_p	Expected Time (t_e)	Variance (σ^2)
1-2	1	1	7	$(1 + 4 + 7)/6 = 2$	$((7-1)/6)^2 = 1$
1-3	1	4	7	$(1 + 16 + 7)/6 = 4$	$((7-1)/6)^2 = 1$
1-4	2	2	8	$(2 + 8 + 8)/6 = 3$	$((8-2)/6)^2 = 1$
2-5	1	1	1	$(1 + 4 + 1)/6 = 1$	$((1-1)/6)^2 = 0$
3-5	2	5	14	$(2 + 20 + 14)/6 = 6$	$((14-2)/6)^2 = 4$
4-6	2	5	8	$(2 + 20 + 8)/6 = 5$	$((8-2)/6)^2 = 1$
5-6	3	6	15	$(3 + 24 + 15)/6 = 7$	$((15-3)/6)^2 = 4$

Let's identify the network paths and their total durations:

- Path 1: $1 \rightarrow 2 \rightarrow 5 \rightarrow 6 = 2 + 1 + 7 = 10$ weeks
- Path 2: $1 \rightarrow 3 \rightarrow 5 \rightarrow 6 = 4 + 6 + 7 = 17$ weeks (Longest Path)
- Path 3: $1 \rightarrow 4 \rightarrow 6 = 3 + 5 = 8$ weeks

i) Expected project completed time:

The expected duration is the length of the critical path (1-3-5-6), which is 17 weeks.

- Project Variance (σ_p^2): Sum of variances on the critical path =
 $1 + 4 + 4 = 9$
- Project Standard Deviation (σ_p): $\sqrt{9} = 3$ weeks

ii) Probability project completed at least 3 weeks earlier than expected:

Target time (X) $\leq 17 - 3 = 14$ weeks.

We find the Z-score:

$$Z = \frac{X - t_e}{\sigma_p} = \frac{14 - 17}{3} = \frac{-3}{3} = -1$$

Using a standard normal distribution table, the probability for $Z \leq -1$ is approximately 0.1587 (or 15.87%).

iii) Probability of not meeting the 18-week due date:

Target due date (D) = 18 weeks. We need the probability that $X > 18$.

First, calculate the Z-score for meeting the deadline:

$$Z = \frac{D - t_e}{\sigma_p} = \frac{18 - 17}{3} \approx 0.333$$

Looking up $Z = 0.33$ in the standard normal table gives a cumulative probability (meeting the deadline) of approximately 0.6293.

Probability of **not** meeting the due date =

$$1 - 0.6293 = \mathbf{0.3707} \text{ (or } \mathbf{37.07\%})$$

Set Number: Set-3

Question COS (a)

Question: Explain with proper analysis of Selective Approach System of control.

Marks: 3

Solution:

The Selective Approach System of Control, most commonly known as **ABC Analysis (Always Better Control)**, is a fundamental inventory management technique. It is based on Pareto's Principle (the 80/20 rule), which posits that a small percentage of items usually accounts for a large percentage of total inventory value.

Rather than applying uniform control to all items—which is inefficient and costly—this system categorizes inventory into three classes based on their Annual Usage Value (Annual Demand $\times$ Unit Cost):

- **A Items (High Value, Tight Control):** These items represent a small fraction of the total inventory quantity (typically 10%–20%) but account for the majority of the inventory value (typically 70%–80%).
 - **Analysis/Control:** Require rigorous control, highly accurate records, frequent review, strict security, and calculation of exact Economic Order Quantities (EOQ). Safety stocks are kept low to avoid tying up capital.

- **B Items (Moderate Value, Moderate Control):** These represent about 20%–30% of inventory items and roughly 15%–20% of the total value.
 - *Analysis/Control:* Require moderate monitoring. Reviews are periodic, and management operates on a management-by-exception basis.
- **C Items (Low Value, Loose Control):** These make up the bulk of inventory items (50%–70%) but contribute very little to the total value (5%–10%).
 - *Analysis/Control:* Require loose control. They are typically ordered in bulk quantities (often 6 months to a year's supply) to minimize ordering costs. Keeping high safety stocks of 'C' items costs very little but prevents disruptive stock-outs.

By analyzing and categorizing inventory this way, a business optimizes its resource allocation, focusing managerial time and capital precisely where it yields the highest financial impact.

Question COS (b)

Question: The XYZ manufacturing company has determined from an analysis of its accounting and production data for part number 625, that its cost to purchase is Rs.36/- per order and Rs. 2/- per part. Its inventory carrying charge is 18% of the average inventory cost. The demand for this part is 10,000 units per annum. What is the economic order quantity and what is the optimal number of days supply per optimum order.

Marks: 3

Solution:

Let's define the given parameters:

- **Annual Demand (D):** 10,000 units/year
- **Ordering Cost (S):** Rs. 36/order
- **Unit Cost (C):** Rs. 2/part

- **Holding Cost (H):** 18% of Unit Cost =
 $0.18 \times 2 = \text{Rs. } 0.36 \text{ per unit/year}$

1. Economic Order Quantity (EOQ):

The EOQ is calculated using the formula:

$$EOQ = \sqrt{\frac{2DS}{H}}$$

$$EOQ = \sqrt{\frac{2 \times 10000 \times 36}{0.36}} = \sqrt{\frac{720000}{0.36}} = \sqrt{2000000} \approx \mathbf{1414.21 \text{ units}}$$

(Practically, the company would order 1414 units).

2. Optimal number of days supply per optimum order:

First, we find the number of orders placed per year (N):

$$N = \frac{D}{EOQ} = \frac{10000}{1414.21} \approx 7.07 \text{ orders/year}$$

Assuming a standard operating year of 365 days, the days supply per order (t) is:

$$t = \frac{365}{N} = \frac{365}{7.07} \approx \mathbf{51.6 \text{ days}}$$

(The company has enough stock from one EOQ order to last approximately 51.6 days).

Question COS (c)

Question: Define strictly deterministic, strictly non-deterministic and fair games. Find the range of values of p and q which will render entry (2, 2) a saddle point for the game.

Pay-off Matrix:

	B1	B2	B3
A1	2	q	6
A2	p	5	10
A3	6	3	4

Marks: $2+5 = 7$

Solution:

1. Definitions:

- **Strictly Deterministic Game:** A game where a saddle point exists. Both players have an optimal pure strategy, meaning they will consistently choose the same single course of action to maximize gains or minimize losses. (Max-min = Min-max).
- **Strictly Non-Deterministic Game:** A game where no saddle point exists. Players must utilize a mixed strategy, assigning a probability distribution over their available choices to optimize their expected outcome. (Max-min $\neq$ Min-max).
- **Fair Game:** A game where the optimal strategies result in a game value exactly equal to zero ($v = 0$). Neither player has a mathematical advantage.

2. Finding the range of p and q :

For the entry (2, 2), which is the value 5, to be a saddle point, it must satisfy two conditions simultaneously:

1. It must be the minimum value in its row (Row 2).
 2. It must be the maximum value in its column (Column 2).
- **Condition 1 (Row Minimum):** In Row A_2 (values: p , 5, 10), 5 must be the minimum. Therefore, p must be greater than or equal to 5.

$$p \geq 5$$

- **Condition 2 (Column Maximum):** In Column B_2 (values: $q, 5, 3$), 5 must be the maximum. Therefore, q must be less than or equal to 5.

$$q \leq 5$$

(Self-Verification: If $p \geq 5$ and $q \leq 5$, the Row Minimums are $[\min(2, q), 5, 3]$ and the Column Maximums are $[\min(2, q), 5, 3]$. The maximum of the row minimums is 5, and the minimum of the column maximums is 5. $\text{Max-min} = \text{Min-max} = 5$. The saddle point condition holds).

Question C06 (a)

Question: In a departmental store, one cashier is there to serve the customers. And the customers pick up their needs by themselves. The arrival rate is 9 customers for every 5 minutes and the cashier can serve 10 customers in 5 minutes. Assuming Poisson arrival rate and exponential distribution for service rate, find:

- Average number of customers in the queue or average queue length.
- Average time a customer spends in the system.
- Average time a customer waits before being served.

Marks: $2+2+2 = 6$

Solution:

This is an M/M/1 queuing model. First, we must standardize the rates to a per-minute basis.

- **Arrival rate (λ):** 9 customers / 5 minutes = 1.8 customers/minute
- **Service rate (μ):** 10 customers / 5 minutes = 2.0 customers/minute
- **Traffic intensity (ρ):** $\frac{\lambda}{\mu} = \frac{1.8}{2.0} = 0.9$

i) Average number of customers in the queue (L_q):

$$L_q = \frac{\rho^2}{1 - \rho} = \frac{(0.9)^2}{1 - 0.9} = \frac{0.81}{0.1} = 8.1 \text{ customers}$$

ii) *Average time a customer spends in the system (W_s):*

This includes both waiting time and service time.

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{2.0 - 1.8} = \frac{1}{0.2} = 5 \text{ minutes}$$

iii) *Average time a customer waits before being served (W_q):*

$$W_q = \frac{\rho}{\mu - \lambda} = \frac{0.9}{2.0 - 1.8} = \frac{0.9}{0.2} = 4.5 \text{ minutes}$$

(Alternatively, $W_q = W_s - \frac{1}{\mu} = 5 - \frac{1}{2.0} = 5 - 0.5 = 4.5 \text{ minutes}$).

Question C06 (b)

Question: What are the advantages of PERT and CPM networks? What is the significance of cost slope?

Marks: 3

Solution:

Advantages of PERT and CPM Networks:

- **Visual Representation:** They provide a clear, logical, and visual flow of project activities and their interdependencies, making complex projects easier to understand.
- **Identifying the Critical Path:** They isolate the sequence of tasks that strictly dictate the project's total duration. Management knows exactly which tasks cannot be delayed.
- **Resource Allocation:** By identifying tasks with "slack" or "float", managers can shift resources (labor, equipment) from non-critical activities to critical ones without extending the project timeline.
- **Time and Cost Estimation:** PERT allows for probabilistic time estimation (handling uncertainty), while CPM allows for evaluating time-cost trade-offs (crashing).

Significance of Cost Slope:

The **Cost Slope** represents the marginal financial cost required to reduce the duration of a specific activity by one unit of time (e.g., one day or week).

$$\text{Cost Slope} = \frac{\text{Crash Cost} - \text{Normal Cost}}{\text{Normal Time} - \text{Crash Time}}$$

Significance: When a project manager needs to shorten a project's duration ("crash" the project), they use the cost slope to determine the most economical way to do so. They will prioritize crashing activities that are on the critical path and have the lowest cost slope, ensuring the project is expedited with the minimum possible increase to the budget.

Question C06 (c)

Question: A company manufacturing plant and equipment for chemical processing is in the process of quoting tender... Help the manager to find the project completion time...

- i) Perform forward and backward pass.
- ii) Identify the critical path and project completion time.
- iii) If activity G is delayed by 2 Days, what happens in project completion time change and/or critical path change?

Marks: 8

Solution:

i) **Forward and Backward Pass Calculations:**

- **Forward Pass:** Earliest Start (ES) + Duration = Earliest Finish (EF)
- **Backward Pass:** Latest Finish (LF) - Duration = Latest Start (LS)
- **Slack:** LS - ES (or LF - EF)

Activity	Duration	Predecessor	ES	EF	LS	LF	Slack
A	3	--	0	3	0	3	0
B	4	--	0	4	4	8	4
C	5	A	3	8	5	10	2
D	6	A	3	9	3	9	0
E	7	C	8	15	10	17	2
F	8	D	9	17	9	17	0
G	9	B	4	13	8	17	4
H	3	E, F, G	17	20	17	20	0

Note on Activity H's Forward Pass: It must wait for E (finishes day 15), F (finishes day 17), and G (finishes day 13). It cannot start until the latest of these finishes, so its $ES = \max(15, 17, 13) = 17$.

ii) Identify the critical path and project completion time:

- The **Project Completion Time** is the maximum Earliest Finish (EF) time of the final activity, which is **20 Days**.*
- The **Critical Path** is the sequence of activities with zero slack. Tracing the zero-slack tasks from the table yields: **A → D → F → H**.*

iii) If activity G is delayed by 2 Days:

*Looking at the table, Activity G has an Earliest Start of day 4 and a Latest Start of day 8. This means it has a **Slack time of 4 days** ($8 - 4 = 4$).*

- A slack of 4 days means Activity G can be delayed by up to 4 days without affecting the start of subsequent activities or the final project deadline.*
- Because a delay of 2 days is strictly less than its available slack of 4 days, there will be **no change to the project completion time** (it remains 20*

days), and there will be no change to the critical path (it remains A-D-F-H).